ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/385104936

Space as Timbre Research-Creation Project Report

Article · July 2024

DOI: 10.5281/zenodo.13621809

CITATIONS
0

READS
49

3 authors:

Pedram Diba
Northwestern University
2 PUBLICATIONS **0** CITATIONS

SEE PROFILE

Jonas Regnier
McGill University
2 PUBLICATIONS **0** CITATIONS

SEE PROFILE

Anqi Liu
University of California San Diego
1 PUBLICATION **0** CITATIONS

SEE PROFILE

All content following this page was uploaded by Pedram Diba on 21 October 2024.

The user has requested enhancement of the downloaded file.

Space as Timbre

Research-Creation Project Report

Pedram Diba, Jonas Regnier, and Anqi Liu

For the original, full version of this publication including scores and sketches, visit:
https://timbreandorchestration.org/writings/project-reports/space-as-timbre

---

1. Introduction 1
2. Collaborative Summary 2
3. Compositional Phase/Structure of the Compositions 6
4. Experience of the Ensemble 9
5. Conclusion 13
References 15
Appendix A 16
Appendix B 19
Appendix C 24

1. Introduction

The primary objective of the Space As Timbre (SAT) project is to conduct meticulous experiments on orchestration solutions and playing techniques, and to achieve timbral effects that closely approximate the perceptual characteristics of diverse acoustical spaces. To this end, SAT endeavors to illuminate the intricate interplay between the spectral content of a target sound, its resynthesis through orchestration, and the resulting perception of the quality of the acoustical space in which the target sound was produced. Presented below is a concise review of a ten-month-long collaboration involving three composers (Pedram Diba, Jonas Regnier, and Anqi Liu) and seven performers (Alex Huyghebaert (flute), Matias Perenetti-Paniagua (clarinet), Éric Bourgeois (trombone), Paul Celebi (piano), Jeanne Côté (violin), William Boivin (double bass), and Charles-Eric Fontaine (conductor)), as well as a sound engineer (Seungwoo Han), from two ACTOR partner institutions, namely McGill University and the University of California San Diego. The collaboration primarily consisted of virtual meetings, workshops, rehearsals, two presentations, a concert, and post-reflections.

SAT's research approach began with the application of various impulse responses[i] to recorded or synthesized sounds, thereby creating a diverse range of acoustical spaces for each sound. Subsequently, these newly formed sounds served as sonic material for instrumental resynthesis. To achieve the instrumental resynthesis, SAT employed target-based computer-assisted orchestration[ii], specifically the software Orchidea[iii], to generate multiple solutions for the orchestration of each target sound. Throughout this process, the primary objective was to explore various settings for partial filtering and sparsity to attain a timbral result that closely matched the

perceptual characteristics of the space of each target sound. In conjunction with the use of Orchidea, the musicians of the ensemble were tasked with critically listening to the audio files and striving to capture the closest perceptual qualities of the space of each target sound by experimenting with a diverse range of playing techniques, including extended techniques, pedaling techniques, bow placement, and more. These experiments were based on the outcomes generated by Orchidea; nonetheless, the musicians were also encouraged to conceive completely new solutions. This collaborative effort ultimately yielded innovative playing techniques that were not suggested by Orchidea.

2. Collaborative Summary

Choice of Target Sounds

Pedram initially started with twelve sounds, which were all unique in nature —sustained sounds, short sounds, sounds with an attack-decay form, grainy and textural sounds, etc., and 402 impulse responses. He explains, "the use of impulse responses can alter what Denis Smalley refers to as the *spectral space* (Smalley, 2007), and consequently the timbre of a sound. As a result, we can achieve various timbral qualities in a sound by changing its space. I experimented with all possible pairings of impulse responses and sounds to find what combinations could create the most interesting results for me." Eventually, Pedram chose two sounds and four impulse responses. The first sound was the recording of a marble bouncing on a balloon that was stretched over the opening of a jar. The second sound was one of the bell recordings included in the Orchidea package called *archeos-bell*. Ultimately, Pedram applied two different impulse responses to each sound and created four target sounds for experimentation with Orchidea and the ensemble - Dry Marble, Marble in Reverberant Space, Bell #1, and Bell #2 .

Similarly, Jonas started his endeavors with a large number of sounds. He explains, "I chose ten different sounds that I wanted to experiment with, and that I found could bring an interesting result after being orchestrated for a small ensemble. Some sounds had textural effects, some had interesting partials or harmonic content, and some had specific gestures." Jonas ultimately chose three sounds. Two of the sounds were created through a digital synthesis software called *Absynth 5*, and the third sound was the sound of a cooking pan hit with a percussion mallet, that he recorded in a sound-proofed studio. "The two synthesized sounds seemed interesting to me not only because of their complex dynamic, harmonic, and timbral characteristics, but because they carried a strong sense of wide-open space within them. I was curious to know whether this spatial effect produced by a digital, artificial timbre could be reproduced with acoustic timbres. The first synthesis sound, called 'Sinus low vibration,' consists of a twelve second low vibration containing a small and progressive glissando (going down by almost half a tone, then up by the same interval). The second synthesized sound 'Vibration F#', is a twenty-five-second slow vibration which is centered around F-sharp. Some of the partials of the sound have different speed and amplitude vibrations." Lastly, Jonas also decided to use the pan sound, which consists of a single attack on a coocking pan with a nine-second reverb time, due to how it would transform when different impulse responses were applied to it. "Although many 'bell-like' sounds have already been orchestrated and used in instrumental music by contemporary composers, I wanted to use this sound nevertheless to try to orchestrate it in different spaces. I

noticed indeed that changing the space (through impulse-response convolution reverb) in which this sound is heard not only changes the timbre of the original sound but also its perceived harmonic content. This is something that I wanted to explore in this piece and used as the main idea for building the form and structure of the piece." The phenomenon of experiencing a distinct shift in perceived harmonic characteristics within a sound following the application of a different impulse response is also clearly apparent in the case of the bell sounds that Pedram specifically chose as his two target sounds: Bell #1 and Bell #2.

In contrast to Pedram and Jonas, Anqi focused solely on one sound: the sound of snow. She intentionally selected this subtle and nuanced sound being aware that it might not align well with the strengths of Orchidea. However, she viewed the ever-changing dynamics and unpredictable nature of the sound of snow as a fitting challenge for her project, believing it would push the boundaries of this computer-aided composition approach. Anqi's primary objective was to investigate how Orchidea would address such an inherently organic and variable sound. Additionally, she aimed to explore the intricate interplay between nature, technology, and human interpretation. "The experience gave me invaluable insights into the balance between the structured precision of technology and the ever-changing dynamics of sonic phenomena. Embracing this tension, understanding it, and allowing it to inform my work has been an immensely rewarding aspect of this journey."

Working with Orchidea

While working with Orchidea in preparation for the first workshop, Pedram used the FullSOL2020 instrumental dataset[iv] to have the software consider a diverse set of playing techniques in the orchestration solutions. With this dataset, he hoped that Orchidea would generate more compelling results in terms of timbre compared to other instrumental datasets such as the TinySOL. "After choosing my dataset, I experimented with the parameters of *sparsity* and *partialsfiltering*[v] to achieve the closest timbral result to each target sound. I also experimented with the parameters of *onsetthreshold* and *onsettimegate* to achieve the best outcome for a dynamic orchestration. If the time gate interval for the onset segmentation was short, Orchidea would generate a lot of material, but the identity of the sound would be lost. If the time gate interval was longer, I would get less material, but the identity of the sound was better preserved in the orchestration solution." During this phase, Pedram attempted to find a fine balance between the four mentioned parameters in Orchidea for each of his target sounds to retain some of their identity features in a dynamic orchestration most important of which for him was the timbral quality. Pedram notes that Orchidea was a very powerful tool for providing orchestration solutions and compositional ideas to him. However, during the process of exploring various orchestration solutions for his target sounds, he also employed his intuition as a composer. "Sometimes, I could hear a timbral quality in a target sound that I was interested in, but not all the solutions from Orchidea would highlight that timbral quality. In those moments, I chose a solution that was timbrally close to the target sound and hoped to explore more solutions with the ensemble during the workshops."

Jonas also employed the FullSol library in his endeavors. However, he initially found working with Orchidea challenging as he thought that textural sounds were not very practical to

orchestrate with the use of Orchidea due to its reliance on the pitch content of the sound. Throughout the orchestration process, Jonas sought to ensure that the most prominent and perceptible musical parameters of the target sounds were also perceptible in the orchestrations, encompassing vibrations, specific rhythms, overall timbre, and space. Guided by this artistic vision, Jonas faced a dilemma between adhering closely to the Orchidea solutions in terms of pitches or preserving the original timbre and rhythms of the sound. "At first, I decided to use all, or almost all the notes given to me by the software in the orchestration solution, but after the first workshop, I realized this was not optimal as the result was not close to the target sound. I then started to get rid of most pitch materials from the solution, and only keep the salient pitches that I could hear from the recording. This gave me more freedom in the use of rhythms and timbre manipulations (through extended techniques), and the results were much closer to the target sounds." Furthermore, during his work with Orchidea Jonas came to the conclusion that the most important parameter for him to experiment with was the *onsettimegate* parameter. "The results varied greatly depending on the value of the time gate, which had to be set accordingly to the target sound: I found that the minimum value for this parameter in order to have an accurate (and not overly complex) solution was 500 ms." Jonas ultimately used values between 1000 ms to 3000 ms for his sounds, as the harmonic content was changing quite rapidly. He notes that he didn't find that the parameters of sparsity and *partialsfiltering* had a notable impact on the orchestration solutions for his specific sounds. He adds that the difference with the orchestration solutions between two similar sounds played in two different spaces was mainly in the partials/pitches. However, he didn't notice any important change in the timbre of the orchestration solution that was generated by Orchidea.

Meanwhile, Anqi, using the TinySol library, grappled with a challenging question during this phase. While Orchidea suggested complex rhythmic patterns and diverse pitch combinations to capture the dynamic essence of her target sound, Anqi questioned the distinction between Orchidea's algorithmic solutions and stochastic choices. This endeavor prompted deep reflection on the nuances lost or overlooked in the computer-aided orchestration process. Anqi emphasized the significance of addressing this issue, given her extensive exploration into the role of meticulously crafted scores, which serve as guiding frameworks for musicians' understanding of their surrounding "spaces" and "ecologies." She remarks, "I perceive past experiences, memories, training, and learnings as integral components of a psychological ecosystem that influence our cognitive patterns, decision-making, and behaviors." Anqi thought that Orchidea did not integrate this human element into its orchestration solutions.

Workshop with the Ensemble

Throughout the course of this project, the composers engaged in three comprehensive workshops alongside the ensemble. Following the conclusion of each workshop, the composers received meticulously annotated parts from the ensemble musicians, providing detailed insights into the discussions and modifications made during the session. With the use of the annotated parts, the composers would refine their material in anticipation of the subsequent workshop, ensuring a continual evolution and enhancement of their compositional material. In preparation for the first workshop, Pedram made several adjustments to the solutions provided by Orchidea. These adjustments encompassed various aspects, such as incorporating glissandi, extending specific

articulations, utilizing arrows to indicate gradual transitions between different playing techniques, accelerating or decelerating certain figures, and employing unison tremolo on adjacent strings. These modifications aimed to create smooth transitions in terms of timbre, pitch, and activity within the different segmentations, while also enhancing the timbral qualities.

Given that the piano was not included in the FullSOL2020 instrumental dataset, Orchidea did not provide any material specifically tailored for the piano. Consequently, Pedram made the decision not to compose anything for the piano for the first workshop. Instead, Pedram invited the pianist to generate suggestions during the workshop, taking into account the target sounds, the sound of the ensemble, and the material provided for the other instruments.

Throughout the three workshop sessions, Pedram, along with the ensemble members, made various modifications to the orchestration solutions. These adjustments encompassed several aspects, including pitch alterations, dynamic markings, and changes to playing techniques, all aimed at getting closer approximations to the timbral qualities of the target sounds. Examples of technique modifications included substituting the half-tone sound for the air sound in the flute, utilizing *molto sul tasto* with half hair in the violin instead of *col legno tratto*, and employing continuous bow bouncing (ricochet) in place of tremolo in the double bass. Furthermore, they delved into the exploration of novel playing techniques, such as performing vocal fry into the bass flute, rubbing a coin on the bass strings of the piano, clarinet key clicks, and simultaneously playing and singing a quarter tone apart in the low register of the flute. These explorations led to the integration of experimental techniques that surpassed the suggestions provided by the FullSOL2020 instrumental dataset in Orchidea. For the final workshop, Pedram revised the orchestration of all the target sounds based on the annotations provided by the ensemble members after the initial two workshops. Furthermore, Pedram endeavored to refine the orchestration solutions further, drawing upon the observations and knowledge acquired during the workshops. For instance, Pedram noticed that incorporating a noise component such as air sound could impart a sense of spaciousness to the timbral quality of a sound. Consequently, Pedram decided to introduce an airy multiphonic in the bass flute for one of the marble sounds, which carried a reverberant and spacious quality. By doing so, Pedram aimed to augment the timbral characteristics of the ensemble's sound, while also creating an overall impression of spaciousness in the sound. During the last workshop, the ensemble experimented with two different multiphonics provided by the composer to determine the optimal execution for achieving this effect.

Jonas had an exceptionally positive experience while interacting with the ensemble. The opportunity to receive direct feedback on the composed musical ideas and their realization proved to be a privilege that Jonas wished to have for the composition of every piece. He notes that the workshops provided valuable insights into orchestration techniques that would not have been attainable within the confines of a traditional orchestration course. Each musician contributed valuable insights regarding the feasibility and effectiveness of the musical materials. In situations where certain aspects of the music did not unfold as intended, the ensemble members offered alternative solutions that bridged the gap between the intended sonority and the actual outcome. The majority of modifications made during the workshops pertained to pitch adjustments and the incorporation of various timbral effects through the utilization of extended

techniques. Since the ensemble musicians had the possibility to collectively listen to the target sounds that the composers aimed to reproduce, they were able to propose innovative solutions to replicate the timbral and spatial characteristics of these sounds, surpassing the scope of Jonas's initial considerations. Additionally, while working with the ensemble, Jonas realized that achieving a unified blend between the instruments was crucial for listeners to perceive the projected space in the music. Without this timbral blend, the intended spatial qualities in the composition would remain largely imperceptible. This experience highlighted the importance of close collaboration and careful orchestration within the framework of the SAT project.

3. Compositional Phase/Structure of the Compositions

Pedram proceeded to create a formal plan for the composition following the workshops. Throughout this process, he gave careful consideration to the pitch organization and the most perceptually prominent pitch in each sound as well as timbral qualities and organizations such as brightness and orchestration. These considerations influenced the decision-making regarding the order of sound appearances and the transitions between them. Pedram's observations ultimately guided him to introduce the marble sounds before the bell sounds in his plan. Additionally, Pedram chose not to commence the piece with the marble sounds, but rather crafted a progression that would eventually lead to the first marble sound, which occurs at the 1:25 mark in the final composition.

During the composition phase, Pedram drew inspiration from Nicolas Donin's article "Sonic Imprints: Instrumental Resynthesis in Contemporary Composition[vi]." He placed particular emphasis on the timbral qualities of the impulse responses and aimed to faithfully recreate those qualities in the final work, rather than precisely reproducing the original sound reference of the target sounds in the composition. Pedram acknowledges that while the instrumental resynthesis does not perfectly reproduce the original sound reference in his piece, familiarity with the source content can aid in recognizing the illusion of the target sounds.

When listening to the premiere performance of the piece, Pedram experienced a dual perception regarding the change of space in the formal structure. Pedram notes that since the target sounds do not appear one after another in the composition, the sense of space between them was challenging to discern. However, the overall change in timbre throughout the larger formal structure of the work was evident. In addition to utilizing the target sounds, Pedram also incorporated various changes of space within smaller formal structures throughout the composition. These changes occurred at a faster pace, making the perception of spatial shifts more noticeable. For instance, in measures 54-57 (second marble sound), the orchestration conveys an open and reverberant space. However, when all instruments drop out in measure 58, leaving the violin alone, Pedram perceived the space to become smaller and drier. Similar examples of spatial changes within smaller formal structures could be found in mm. 1-4 or mm. 4-13, where the progressively brighter sound gives the impression of the source getting closer in space.

Jonas decided to incorporate inspiration from extra musical elements during the composition phase. He created the general form of the piece based on three different light techniques used in

photography and their metaphorical spatial meaning. The first section of the composition, spanning from the beginning to measure 55, incorporated two distinct photography techniques. The first technique, known as "tension," aims to generate contrast in photographs with the use of light. In the context of his piece, Jonas tried to use space to create opposition and tension within various elements. The second technique, called "emphasis," directed the listener's attention to the crucial aspects of the composition through the use of lines, shapes, and negative space, which refers to the area surrounding an object. A negative space usually doesn't contain as many details as a positive space. This concept is similar to the concept of background and foreground in music. In the context of the work this form of opposition is derived from the dichotomy between the initial attack (m. 1) and its composed resonance, both of which come from the orchestrated sound of the pan. Jonas also tried to further evoke tension in this section by breaking the expectation of the listener (via the use of silences), "blurry" effects (small modifications of musical parameters in materials that have previously appeared in the piece), and by progressively building up the polyphony. The attack and resonance motifs reappeared in increasingly complex forms, ultimately merging to create a completely new sonic space (mm.39-55). This newly introduced space maintains the essence of tension and opposition present at the beginning of the piece. However, in this instance, the contrasting materials are juxtaposed simultaneously, rather than sequentially. The foreground features frantic, chaotic, and noisy elements, which serve as a counterpoint to the background's persistent ascending melody played on the piano. The piano melody oscillates in dynamics while overall gradually increasing in volume, progressively emerging into the foreground. This interplay of contrasting elements within the space exemplified the concept of negative space, as listeners' attention alternates between the chaotic noise and the piano melody, with the latter continually shifting between foreground and background.

In the second section of the piece, spanning from measure 56 to 84, Jonas explored the concept of "substance" derived from photography. This technique aims to enhance the depth of the narrative by reinforcing the primary subject through the use of light or space within the concept of Jonas's composition. The section begins by prominently featuring the "Vibration F#" sound, introducing a second element to the narration while concurrently altering the spatial environment. At measure 69, a dramatic clash between registers occurs on the piano, contrasting a loud chord utilizing extreme registers with a quiet chord confined to a restricted register. This clash generated a sudden shift in space, transitioning from a wide space to a more narrow one. The transformation in space also includes the reminiscence of the resonance of the pan sound previously heard in the first section of the piece. By the conclusion of the section, the narrow space material grows louder and precedes the wide space material.

The final section of the composition, spanning from measure 85 until the end, deals with the photography technique known as "separation." This technique aims to shape the narrative by employing basic tonal differentiation. The section begins in silence and gradually expands the space through the accumulation of rhythms, melodic lines, harmonies, and timbres. At the beginning of this segment, a new musical element emerges, partially introduced in the concluding measures of the second section (m.79), starting with the violin part and subsequently integrating with other instrumental parts. This element consists of a recurring descending scale that grows progressively longer, softer, and lower in pitch. The unfolding of this heterophonic

texture gradually opens up a wide space, subsequently filled by the final musical material of the piece, namely the "Sinus low vibration" sound. This concluding sound slowly fades away within the same space it had initially occupied, leaving behind a quiet and never-ending resonance as an integral part of the space.

In contrast to Pedram's meticulous attention to pitch organization and timbral qualities, and Jonas's incorporation of extra-musical elements, Anqi's exploration diverged after the workshops as she delved into the realm of musical notation for her compositional process. She explains, “Instead of solely relying on computer-aided software, I felt an urge to explore alternative solutions. This divergence wasn't just about moving away from technological means, but it also called into question the very linguistic model that has anchored musical endeavors for centuries: notation.”

In her reflection, Anqi explored the realm of linguistic models, pondering their role in simplifying intricate phenomena into language that is universally comprehensible. This contemplation led her to draw a parallel between linguistic models and traditional music notation, recognizing the latter as a framework for condensing expansive sonic landscapes into simple symbols on staves. She further elaborates, “Just as technology provides a framework to process and produce, so does musical notation offer a language for our sonic expressions. Both, at their heart, are linguistic models, shaping and codifying our interactions with sound.”

However, Anqi acknowledges the limitations of notation, recognizing that every symbol carries cultural, historical, and personal baggage, potentially hindering its ability to fully capture the essence of sound or the musician's intended emotion. Her reflection ultimately guided her to seek a deeper understanding of human connection, gesture, and emotion in her work.

In light of these revelations, she sought to carve out microscopic spaces within the confines of musical notation. While acknowledging the richness of a traditional score, she recognized its limitations in its linear, time-bound representation of sound. Her endeavor aimed to transcend these constraints, offering a more holistic, spatial, and interactive experience. She sought to create a dynamic system where notation itself would evolve, akin to the ecologies previously mentioned. This new notation would integrate the performer's experiences, memories, and training, resulting in a transient rendition with each performance, thereby bridging the gap between the rigidity of linguistic models and the fluidity of human experiences.

Consequently, she introduced the concept of threshold in her notation to signify a state that is in-between and ongoing. She explains, “In this situation, uncertainties and unpredictabilities are unenviable. The thresholds are revealed in multiple parameters while playing the instruments. “Part of the concepts of the piece is to find, explore, and define those thresholds. The thresholds are conditions of uncertainties and are conditions of the edge of new or different things starting emerging; therefore it’s always in the in-between and either-or situations.” An example of this can be observed in the strings shown in Figure 1, where thresholds are applied to the parameters of the bow. As the bow interacts with the string, factors such as its positioning and pressure serve as variables, influencing the pitch, volume, and overall sound quality produced. Despite the complexity of these variables, they remain limited by the constraints of physicality.

Nevertheless, it is precisely this physical aspect that facilitates the expression of the abstract. Anqi elaborates, “my scoring system embraces the unpredictable, allowing it to flow freely, shaped by a multitude of human phenomena. The essence of the score changes when each performer brings their personal touch, be it through subtle differences in interpretation, fleeting moments of subconscious interference, or even a rekindling of past memories evoked during the performance. These moments, rich with individual experience and unspoken narratives, can influence the micro gestures utilized to approach these thresholds. Instead of being precise and defined, the application of technique becomes a reflection of these underlying psychological ecologies.”

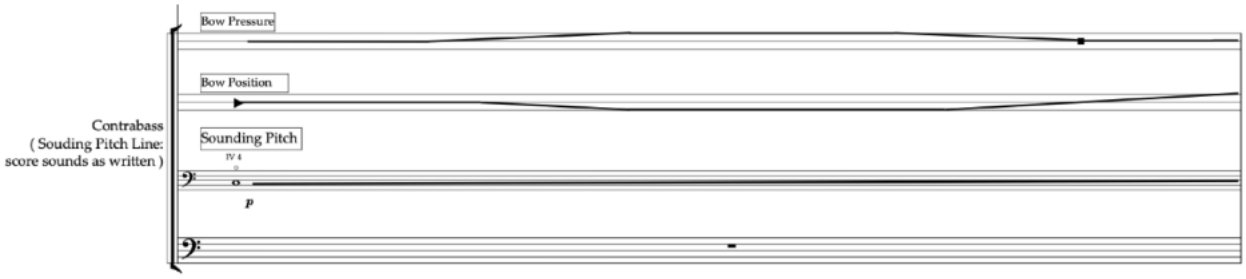


Figure 1: Example of threshold in bow placement and pressure on the double bass

Anqi explains that this approach to orchestration and notation aims not to replace traditional systems, but to enhance them. Through these new systems and ongoing exploration, Anqi’s goal is to create a space where technology complements human experiences rather than restricts them. Here, linguistic models provide foundational frameworks that are flexible enough to accommodate the unpredictable nature of human experiences. This journey embodies Anqi’s philosophical reflections on space, ecology, human connection, and the delicate balance between structure and uncertainties.

4. Experience of the Ensemble

Following the conclusion of the concert, we sought the insights and perspectives of the musicians comprising the ensemble in order to capture their experiences throughout the entirety of this project. Presented below is an assortment of carefully curated questions tailored specifically for the performers, alongside some of their respective reflective responses. Although there were seven performers in the ensemble, the responses included here are from four of the musicians, chosen to provide a representative sample of the performers' diverse perspectives and experiences throughout the SAT project.

What did you find the most fascinating during this project in relation to timbre and space?

Alex Huyghebaert (flute): I was very fascinated by how easily we were able to improve the sound from the original output generated by Orchidea. Just using our ears we were able to make better decisions on which harmonics to bring out and make better choices on dynamics to imitate the target sound. Because computers can pick up more pitches and sounds than human ears

notice, Ochidea would notate these pitches, but we couldn't actually hear them in the original sound.

I was also kind of surprised by the fact that I never thought about how the space that we played in as the ensemble would affect the sound until the day of the concert in Tanna Hall.

Éric Bourgeois (clarinet): I got to pay more attention to low and high frequencies to accurately reproduce the sounds in different spaces.

Jeanne Côté (violin): The most fascinating thing about this project is the way it led me to develop a better sensibility to space as a performer and a listener. The more details I was able to identify on one sound (for example, the length, the harmonics, the attack.), the more I was able to understand the orchestration and to play with the right intention.

Charles-Eric Fontaine (conductor): To me, the reflection with the composers and performers about finding musical solutions to alter metaphorically the perception of space was the most fascinating aspect of this research. As performers, we are used to adjusting details of playing (i.e. length of notes, dynamics, articulations) to adapt to the acoustical properties of a hall or space. During this research, the goal was to implement sound objects with a set of variations to give them the illusion that they are ringing in a different environment. We came to a vast pool of ideas about how to manipulate these sounds: their voicing and balance of the voices among instruments, presence and absence of a specific spectrum of overtones, quality of articulation and bowing, specificity of sound production on extended techniques, and so on.

How did you approach thinking of spatial quality on your instrument and how would you define the role of your own instrument within the orchestration of the target sounds we studied, among the 5 other instruments?

Alex Huyghebaert (flute): I've always thought of the flute, even in a classical orchestral setting, as the top harmonic of sound. It's easier to tune when thinking of it this way - to fit into the sound of the lower orchestra. I think flute was used in this way for SAT maybe a couple times but honestly not as much as I had thought. It helped obviously give the sound depth because of its register in comparison to trombone or bass, but my role kind of changed with each target sound. Sometimes my role was as I said - being a harmonic or higher partial - but sometimes I was part of the main meat of the sound helping change the colour (to be brighter or more airy), sometimes my role was to fill out notes in a chord (i.e. for harmonic structure), and sometimes I created texture or rhythm (like the marbles dropping + vocal fry). It was interesting to see how my biases took me always towards thinking of flute as the higher partial, whereas others would suggest another role for me and what ended up happening was something different. So mostly I'd say I was either a higher partial, a modifier of colour, a contributor to harmony, creator of textures, or equal with the rest of the ensemble.

Éric Bourgeois (clarinet): I found the trombone had to play the role of a bass instrument, along with the actual double bass we had with us, but also the trombone played a role in the mid-range frequencies to fill in the gaps between the low and high instruments and make the sound thicker.

Jeanne Côté (violin): In my opinion, thinking about space when playing the violin boils down not only to working on the timbre of the instrument in a given acoustic, but also to going beyond the properties of this space. This was done by exploring different playing techniques with the ensemble. Through the experimentation sessions, we explored several combinations. Due to the range of my instrument and my timbre palette, the sounds of high harmonics and incisive attacks were often attributed to me.

How did you experience the space parameter while playing a fraction of the sum total of the sound?

Alex Huyghebaert (flute): Like any music I guess. You have to know your role in the overall timbre just as you would if you were playing Beethoven. Except in this case we had a sound to imitate, whereas Beethoven is... Beethoven. The goal most of the time with music like Beethoven is to move the audience emotionally or impress them with virtuosity. The purpose of SAT particularly at the beginning was to recreate a sound we had just heard, which had its own challenges but in some ways was even actually more straightforward. So my answer would be....I experienced it exactly as I would experience it in any other context, with any other music, only that the goal timbre and sound had a different aesthetic purpose.

Éric Bourgeois (clarinet): I had the tendency to hear myself a lot more than I wanted to, but it was a good exercise to try to really blend in with the rest of the group and not come out on top.

Jeanne Côté (violin): Within the ensemble, it is difficult to hear the space. The physical position occupied by the musicians does not allow for an accurate representation of the sound of the whole. Despite everything, certain composition and interpretation techniques made me hear the space. This is the case of variations in playing techniques, articulations and volume balance between instruments.

How did taking part in the workshops influence the way you were performing the pieces at the concert?

Alex Huyghebaert (flute): It gave so much more context on what our sound goal was. If there was an excerpt in the piece that came from the workshops it was easy to connect to the other instruments, understand my role, and understand the point and musical goal of the score. It also helped morale between musicians I think as well; I really respected everything that was suggested during the workshops and you really got to see everyone's creativity and personality coming out when problem solving, and it gave so much more depth to my understanding of my fellow musicians as creators and how I could work and create with them in the moment of performance.

Éric Bourgeois (clarinet): I felt some moments throughout the pieces were meant to be played with close attention to the sound, which reminded me of the workshops.

Jeanne Côté (violin): The experiments preceding the creation of new pieces allowed me to understand the material constituting the compositions. I was able to get to know the work and

style of Anqi, Jonas and Pedram, including their different perception of space. My interpretation was greatly influenced by these moments of exchange.

Charles-Eric Fontaine (conductor): The workshops were aiming at listening to target sounds on which we implemented impulse responses, comparing them to the orchestration generated with Orchidea, and then bringing ideas in from composers, performers and conductor altogether to enhance these orchestrations. As performers were put in contact with these sound materials very early on in the process towards the concert, as well as reflecting and troubleshooting the proposed orchestrations, I perceived that the result at the concert was that the ensemble played in a cohesive and confident manner. Indeed, through this process, each performer had listened over and over to the target sounds and had reflected upon its role in the orchestration of these materials. Therefore, our collective goal was absolutely clear and every performance detail we implemented took root in that experience of the workshops. It was a very effective process and a rewarding performance from my point of view.

What's still on your mind now that this project is over? Would you like to share anything else of your experience?

Alex Huyghebaert (flute): It was just such an amazing experience using our own ears to mimic another sound and be successful at it. Listening back to our recordings was just so encouraging as a musician.

Éric Bourgeois (clarinet): Creating music of this genre is not easy. I learned that it requires attention to very small details, a lot of patience to put in the work, and open mindedness to actually take the music on paper and really try to produce a piece out of it because it doesn't happen on its own.

Jeanne Côté (violin): As a performer, I think listening to the space is an interesting aspect to develop. Indeed, the disadvantageous physical position of the musician makes this listening difficult to apply. I would like to add another thought. The interpretation of a new composition already requires great concentration. The space available for listening to space is therefore limited. With practice, I think it would be possible to develop this skill further.

Charles-Eric Fontaine (conductor): I consider this research project as the exploration phase for a larger-scale project about the component of space perception in music. It was a fascinating process, and I hope to be able to pursue the orchestration experimentation and theorization of the spatial dimension from the compositional and performative angles.

Has this process influenced your way of listening and/or practice? If yes, how?

Alex Huyghebaert (flute): It has made me question a little bit my role as a flute player. I don't always have to be the high voice, the solo melody, the little cherry on top of the C major chord. I can have other roles and purposes in an ensemble.

Éric Bourgeois (clarinet): Yes, I pay more attention to the different elements that might constitute a sound, just for the fun of it. The workshops did teach me a little bit about the art of production.

Jeanne Côté (violin): Following this project, I try to integrate a listening of space into my violin practice and thus adapt my playing according to my needs. I manage to apply this practice only when I have a good knowledge of the score.

Charles-Eric Fontaine (conductor): This research definitely influenced the way I relate to sounds in rehearsal, and how I listen to music in general. Since then, I am more sensitive to conceptualizing certain sound objects as spatial structures and in dialogue with their spatial resonance. I developed a personal toolbox with some of the "spatial manipulations" of the sounds we workshopped, which I use when studying and understanding scores, but also when rehearsing to deepen the interpretative and performance components. It changed my ear to the degree that I now understand better certain composers and genres.

What are some examples of some new playing techniques you discovered on your instrument?

Alex Huyghebaert (flute): Vocal fry into the instrument. This was one that I came up with while trying to imitate the rapid bouncing of marbles. And it made it into the final piece! I was so proud hahah

I think it's a really cool effect because it sounds odd, the audience can't really tell how the musician is making the sound, and it's kind of creepy as well, very unknown.

Éric Bourgeois (clarinet): I did discover how to “slap tongue” effectively and loud, something I didn’t think was possible.

Jeanne Côté (violin): I don't think I discovered any new violin techniques during this project.

5. Conclusion

Before the emergence of electronic music, the sound of an ensemble was bound to the placement of the musicians and the space in which they performed. Nonetheless, the exploration of spatial elements within instrumental music is not a novel concept, as numerous composers have ventured into spatial experimentation, resulting in diverse outcomes regarding the perception of distance, depth, and movement in their compositions. For example, Giovanni Gabrieli used stereo choirs in St Mark’s Basilica in Venice, and Berlioz created the perception of distance through various wind instruments in his Requiem (Normandeau, 2009). Other composers have also sought to utilize space by spatializing sound sources or timbral sources within the performance space using instruments. One compelling example is Philippe Leroux's composition *De la Texture*, which provides specific instructions to performers to navigate designated pathways while executing the piece. Although this method effectively spatializes various sounds

and timbres, the spatial component in such a work is primarily perceivable when experiencing it in a live performance. The recorded version of the work tends to lose the full spatial essence.

Conversely, other composers who have explored the concept of space may have encountered contrasting outcomes. An illustration of this is Pedram Diba's *As Close As Breath*, a collaborative work with violinist Jeanne Côté, developed as part of the Musician's Auditory Perception (MAP) research project with ACTOR. In the final recording of the composition, Diba and Côté utilized MS-EHB-2-Sound Professionals-ultra low noise ear-hook binaural microphones. The use of binaural microphones facilitated the perception of sound movement within the intimate sonic space of the performer, as well as enhancing the quality of the intimate space, particularly during moments involving breath sounds or when the performer brushes the violin against her hair. However, the live performance of the piece loses much of its spatial quality due to the impracticality of employing binaural microphones, which leads to feedback issues, necessitating the use of DPA microphones instead.

The efforts undertaken within the framework of Space As Timbre lay the foundation for innovative approaches to incorporating space in music composition and performance. By considering space as a timbral effect, there is an opportunity to integrate spatial elements into composition in a manner that preserves the timbral structures and trajectories in both live performances and recorded versions of a piece. Moreover, the discovery of techniques that go beyond Orchidea's suggestions, such as playing vocal fry into the bass flute, creates an opportunity to develop new datasets for Orchidea to expand its orchestration capabilities.

In the future, it would be intriguing to conduct an in-depth analysis of the compositions developed within this project, aiming to identify similar instances of playing techniques or sounds that contribute to the perceptual qualities of various acoustical spaces. For instance, in both Pedram's "marble sound" and Jonas's "pan sound" we can observe that instrumental synthesis necessitated the inclusion of noise components (airy sound) when the orchestration aimed to depict a spacious and reverberant environment. Other intriguing future endeavors could include exploring how the perception of projected space in music necessitates a unified blend between instruments, as realized by Jonas during his work, and analyzing the different orchestration techniques used to create this blend in the three pieces written within the framework of this project. Additionally, exploring how traditional versus innovative notation methods, as examined by Anqi, might alter the outcome would be an interesting future endeavor. Lastly, the perceptual qualities of space extend beyond the mere reverberation and spaciousness of a given environment. Parameters such as delays, grains, and more also influence our perception of space. Exploring such factors in future research endeavors would be worthwhile.

Program Notes

Different Same
This work explores how the space we put a sound in can affect how we perceive that sound. Poetically, we can also see the influence of space in our lives: various aspects of our environment and the space we are put in, such as family, home country, hometown, historical

and political aspects, socioeconomic status, language, schools, etc., can directly impact how we perceive our world.

Flou, Flash, Lucide
With this piece, I intended to make a metaphor between photography and music, by drawing an analogy between light as used in photography, and space in music. Both can be used to create tension, emphasis on small details, a sense of separation or even to provide additional depth to the main materials. I was inspired from photography and light techniques, which allowed me to develop orchestration techniques that would directly influence our perception of space. Just like in photography, some musical materials in the piece can be clearly identified while others may be perceived as blurry and harder to grasp.

While Snow Buries…
Over the past years, I have been working mainly on fragile nuances embedded in the sonic phenomenon. These practices always involve slow time evolution for the nuances to be perceived. I thus wished to explore space with chaotic textures in the "Space As Timbre" research project. The feature of Orchidea (the computer- aided patch) seems not supportive of this kind, but it helps me greatly in a way to determine and continue what's important to me. The composition didn't use Orchidea nor other computer-aided software. It is an expansion of the Etude Series that the first etude was composed back in 2019 for the ACTOR CORE seminar at UCSD. These etudes view performers' micro-kinetic capabilities and psychological thresholds at the edge of their bodily and mental ecologies and keep questioning the limits of that edge; function notated score as heuristic space that encourages the performers into self-exploration. They then can observe, experience, and finally break previous boundaries.

References

- Bampis, Evripidis, et al. (2023)."Target-based Computer-assisted Orchestration: Complexity and Approximation Algorithms." *European Journal of Operational Research* 304, no. 3: 926-938.
- Cella, Carmine-Emanuele.(2021). *Workshop on Computer-assisted orchestration and Orchidea*. Ircam, streaming video. YouTube.
- Diba, Pedram. (2021). *As Close As Breath.* Paris: BabelScores. BabelScores catalogue.
- Donin, Nicolas. (2015). “Sonic Imprints: Instrumental Resynthesis in Contemporary Composition.” In *Musical Listening in the Age of Technological Reproduction*, 323–44. Farnham: Ashgate.
- Ircam, HEM, UC Berkeley, and Carmine-Emanuele Cella. "Orchidea: Creative Computer-Assisted Orchestration." Accessed June 11, 2024. http://www.orch-idea.org/.
- Leroux, Philippe. (2008). *De la texture*. Paris: Gerard Billaudot.
- Leroux, Philippe. (2014) *De la texture.* Meitar Ensemble. Pierre-Andre Valade. Meitar Ensemble, streaming video. YouTube: https://www.youtube.com/watch?v=EwTFRQ15bpA
- Normandeau, Robert. (2009). “Timbre Spatialisation: The Medium Is the Space.” *Organised Sound: An International Journal of Music Technology* 14, no. 3 : 277–85. RILM.

- Schneider, Berk, Florian Grond, Jeanne Côté, Pedram Diba, Min Seok Peter Ko, Sang Song, Tiange Zhou, and Shahrokh Yadegari. (2022). "Musicians Auditory Perception: Listening and Empathizing in the Creative Process." *Timbre and Orchestration Resource: Interactive Project Report*: 1-38. ACTOR Project.
- Smalley, Denis. (2007). "Space-Form and the Acousmatic Image." *Organised Sound: An International Journal of Music Technology* 12, no. 1: 35–58. RILM.
- The Timbre and Orchestration Resource. (2024). *Orchidea Tutorials*. https://timbreandorchestration.org/all-videos/tutorials/orchidea-tutorials?rq=tutorial, Accessed June 11, 2024.

Appendix A

The Creation and Realization Process for Different Same Individual Report

Pedram Diba (Northwestern University)

My goal during the Space As Timbre project was to use Orchidea, a computer-assisted orchestration package in MAX MSP, and the creativity and practical knowledge of instrumentalists to recreate - with the medium of an acoustic ensemble - the timbral effects of various acoustic spaces applied to particular target sounds. The application of acoustic environments onto the sounds was achieved with the employment of various impulse responses. The use of impulse responses can alter what Denis Smalley refers to as the spectral space (Smalley 2007), and consequently the timbre of a sound. As a result, we can achieve various timbral qualities in a sound by changing its space.

Initially, I started with twelve sounds, which were all unique in nature - sustained sounds, short sounds, sounds with an attack-decay form, grainy and textural sounds, etc. - and 402 impulse responses. I experimented with all possible pairings of impulse responses and sounds to find what combinations could create the most interesting results for me. Eventually, I chose two sounds and four impulse responses. The first sound was the recording of a marble bouncing on a balloon that was stretched over the opening of a jar. The second sound was the bell recording that was available in the Orchidea package. Ultimately, I was able to apply two different impulse responses to each sound and create four target sounds for experimentation with Orchidea and the ensemble.

While working with Orchidea in preparation for the first workshop, I used the *FullSOL2020* instrumental dataset to have Orchidea consider a diverse set of playing techniques in the orchestration solutions. With this dataset, Orchidea generates more compelling results in terms of timbre compared to other instrumental datasets such as the *tinySOL*. After choosing my dataset, I experimented with the parameters of *sparsity* and *partialsfiltering* to achieve the closest timbral result to the target sound. I also experimented with the parameters of *onsetthreshold* and *onsettimegate* to achieve the best outcome for a dynamic orchestration. If the time gate interval

for the onset segmentation was short, Orchidea would generate a lot of material, but the identity of the sound would be lost. If the time gate interval was longer, I would get less material, but the identity of the sound was better preserved in the orchestration solution. I tried to find a fine balance between the four mentioned parameters in Orchidea for each of my target sounds to retain some of their identity features in a dynamic orchestration most important of which was the timbral quality.

Orchidea was a very powerful tool for providing orchestration solutions and compositional ideas to me; however, I also employed my intuition as a composer during the process. For example, sometimes I could hear a timbral quality in a target sound that I was interested in, but not all the solutions from Orchidea would highlight that timbral quality. In those moments, I chose a solution that was timbrally close to the target sound and hoped to explore more solutions with the ensemble during the workshops.

In preparation for the first workshop, I made some adjustments to the solutions provided by Orchidea. Some of these adjustments included adding glissandi, continuing particular articulations, adding arrows to indicate a gradual shift from one form of playing to another, accelerating and decelerating some figures, unison tremolo on two adjacent strings, etc. All these modifications served to create more smooth transitions in terms of timbre, pitch, and activity between the different segmentations as well as providing timbral augmentation. Since the piano is not included in the FullSOL2020 instrumental dataset, Orchidea did not provide any material for the piano. For the first workshop, I decided not to write anything for the piano either. I asked the pianist to generate some suggestions during the workshop while listening to the target sounds and the ensemble and reading the solution that was provided for the rest of the instruments.

Throughout the three workshop sessions, I made various modifications to the orchestration solutions with the help of the ensemble members. Some of these modifications included changing some of the pitches and dynamic markings as well as altering some playing techniques to get closer to the timbral quality of the target sounds. Some of the technique alterations included using half-tone sound instead of air sound in the flute, molto sul tasto with half hair in the violin instead of CLT, and continuous bouncing of the bow (ricochet) instead of tremolo in the double bass. We also explored new playing techniques such as performing vocal fry into the bass flute, rubbing a coin on the bass strings of the piano, clarinet key clicks, and simultaneously playing and singing a quarter tone apart in the low register of the flute. Exploration of new techniques helped us find some experimental techniques that even the *FullSOL2020* instrumental dataset in Orchidea could not have suggested.

For the last workshop, I edited the orchestration of all my target sounds based on the annotations provided by the ensemble members after the first two workshops. I also tried to edit the orchestration solutions further based on what I had already observed and learned during the workshops. For example, I noticed that the presence of air sound can create a sense of spaciousness in the timbral quality of a sound. Therefore, I decided to add an airy multiphonic in the bass flute for one of the marble sounds that had a very reverberant and spacious quality. By doing so, I hoped to provide timbral augmentation based on what was taking place in other instruments and create an overall sense of spaciousness in the sound. During the third workshop

we experimented with two different multiphonics to find the best option for the execution of this effect.

After the workshops, I created a formal plan for my piece. During this process, I had to consider - in each sound - the pitch organization and the central pitch, timbral qualities such as brightness, orchestration, etc., to decide in which order I wanted the sounds to appear as well as how the music would continue from one sound to the next. Based on my observations, I planned to have the marble sounds first, followed by the bell sounds. Additionally, I decided not to start the piece with the marble sounds, but rather compose my way to the first marble sound, which in the final work appears at minute 1:25.

During the composition phase, I focused on the timbral qualities of the impulse responses and tried to recreate those qualities in the final work. In his article "*Sonic Imprints: Instrumental Resynthesis in Contemporary Composition*" Nicolas Donin talks about the notion of referential dimension and identity in a sound and how the process of resynthesis can affect them. He argues that regardless of what the sound referent might be, the illusion may not be perfect enough for the listener to recognize the reference. Despite this, source content awareness can help a listener perceive the illusion of the referential dimension of a resynthesized sound. Resynthesis can also affect the identity of a sound. However, a sound can retain certain essential features that define its identity even if identity is not the primary concern through the resynthesis process (Donin 2015). During the composition phase, I was not concerned with the referential dimension of the target sounds in my piece, but rather their timbral identities, which resulted from the application of the impulse responses. Nonetheless, familiarity with the source content can help one recognize the target sounds in *Different Same.*

While listening to the premiere performance of the piece, I both noticed and didn't notice the change of space in the formal structure of the work. Since the target sounds do not appear one after another in the composition, it was difficult to feel a change of space between them. However, the overall change of timbre within the large formal structure of the work was clear to me. In addition to using the target sounds, I also attempted to create various changes of space in small formal structures throughout the work. Since these changes take place much faster, the perception of change in space is more noticeable. For example, in mm. 54-57 (second marble sound) the orchestration portrays an open and reverberant space. However, once all the instruments drop out in m. 58 and the violin continues by itself, I perceive the space to become much smaller and dryer. Other similar examples in the small formal structure are mm. 1-4 or mm. 4-13 where the sound keeps getting brighter, giving a sense that the source is getting closer in space.

Overall, I found this experience to be very inspiring, and I learned a great deal about approaching space in instrumental music. I found the method of using the timbral qualities of sounds in different spaces to create the formal structure of a piece to be effective for me. Additionally, it was reassuring to see that on occasions where the time between two events with different spatial qualities was short, the change of space could be noticeable. For future works, I would be interested to incorporate the change of spatial qualities in a more connected manner between the large formal structure and the small formal structures.

References

- Bachelard, Gaston. (1969). *The Poetics of Space*. Boston: Beacon Press.
- Cella, Carmine Emanuele. (2012). *Workshop on Computer-assisted orchestration and Orchidea*. Ircam, streaming video. YouTube.
- Donin, Nicolas.(2015). "Sonic Imprints: Instrumental Resynthesis in Contemporary Composition." In *Musical Listening in the Age of Technological Reproduction*, 323–44. Farnham: Ashgate.
- Smalley, Denis. (2007). "Space-Form and the Acousmatic Image." *Organised Sound: An International Journal of Music Technology* 12, no. 1: 35–58.

Appendix B

Individual Project Report

Jonas Regnier (McGill University)

I. Project Introduction

The ACTOR research-creation project Space as Timbre aims to use timbre as a main parameter to evoke different spaces in a sound through orchestration, and endeavours to treat space as a form- bearing parameter in music. As a composer involved in this project, I composed a 6-minute piece entitled *Flou, flash, lucide.*

This project involved three different phases for composers (each phase lasting a few months): the first phase (research phase) consisted in defining what the term space implies and means, finding some recordings (target sounds) to orchestrate, varying the perceived space of these recorded sounds by modifying the impulse-response parameter with a convolution-reverb software (MaxMSP) and learning to use the orchestration software Orchidea to recreate an instrumental version of the target sounds.

The second phase (workshop and experimentation phase) consisted in having the orchestrated target sounds performed and rehearsed by the ensemble (bass flute, Bb clarinet, trombone, piano, violin and double bass). This phase allowed a direct collaboration between the composers and the performers, as the performers were giving feedback and advice on the different extended techniques they could play on their instrument as well as on the general orchestration of the target sounds.

The final phase (composition phase) of the project consisted of writing a 5-to-7-minute piece that used all the orchestrated target sounds (while taking into consideration the feedback received during the workshops) and treated space as one of its main parameters.

II. Composition Process

a) Choice of target sounds

At first, I chose ten different sounds that I wanted to experiment with, and that I found could bring an interesting result after being orchestrated for a small ensemble. Some sounds had textural effects, some had interesting partials or harmonic content, some had specific gestures. For reasons stated below, I then decided to focus on three different sounds. two sounds were created artificially through a digital synthesis software called *Absynth 5*, and the last sound is the sound of a cooking pan when hit with a percussion mallet, that I recorded in a sound-proofed studio a few years ago.

The two synthesized sounds seemed interesting to me not only because of their complex dynamic, harmonic and timbral characteristics, but because they carried a strong sense of wide-open space within them. I was curious to know whether this spatial effect produced by a digital/artificial timbre could be reproduced with acoustic timbres. The first synthesis sound, called "Sinus low vibration," consists of a twelve second low vibration containing a small and progressive glissando (going down by almost half a tone, then up by the same interval). The second synthesis sound "Vibration F#," is a twenty-five-second slow vibration which is centered around F-sharp. Some of the partials of the sound have different speed and amplitude vibrations.

The recorded pan sound, which consists of a single attack on the pan with a nine-second reverb time, seemed interesting for a different reason than the two synthesized sounds: although many "bell-like" sounds have already been orchestrated and used in instrumental music by contemporary composers, I wanted to use this sound nevertheless to try to orchestrate it in different spaces. I noticed indeed that changing the space (through impulse/response convolution reverb) in which this sound gets played not only changes the timbre of the original sound but also its perceived harmonic content. This is something that I wanted to explore in this piece and used as the main idea for building the form and structure of the piece.

b) Orchestration of target sounds with Orchidea

At first, I was finding it challenging to understand how to make the most out of Orchidea, as it was quite difficult to get an orchestration that was close to the target sound, even when using the FullSol library. I quickly realized that textural sounds were not very practical to orchestrate with Orchidea as the software relies too much on the pitch materials of the sound (instead of timbre). I also realized that the whole orchestration process was going to take a very long time: firstly, to have Orchidea find a satisfying orchestration solution, and then to transcribe it on the score with the appropriate time signature, rhythms, polyphony, timbres, extended techniques, etc. I found myself in a dilemma between staying close to the Orchidea solution (in terms of pitches) or staying close to the original timbre and rhythms of the sound. At first, I decided to use all, or almost all the notes given to me by the software in the orchestration solution, but after the first workshop, I realized this was not optimal as the result was not close to the target sound. I then started to get rid of most pitch materials from the solution, and only keep the salient pitches that I could hear from the recording. This gave me more freedom in the use of rhythms and timbre

manipulations (through extended techniques), and the results were much closer to the target sounds. I would say that the exact imitation of the target sounds was slightly more important than keeping the musical parameters offered by the sound (especially the different pitches given by Orchidea), but I still made sure that the most salient perceptible musical parameters we perceived of to the target sounds were also perceived in the orchestrated sounds (vibrations, specific rhythms, overall timbre and of course, space).

During my experimentations with Orchidea before the workshops, I found that the most important parameter to experiment with was the "onetimegate" parameter. The results varied greatly depending on the value of the time gate, which had to be set accordingly to the target sound: I found that the minimum value for this parameter in order to have an accurate (and not overly complex) solution was 500 ms. I used values between 1000 ms to 3000 ms for my sounds, as the harmonic content was changing quite rapidly. I didn't find that the other two parameters (sparsity and partial filtering) had a notable impact on the orchestration solutions. The difference with the orchestration solutions between two similar sounds played in two different spaces was the partials/pitches. I didn't notice any important change in the timbre that was given by Orchidea in the solutions (despite using the FullSol library).

c) Orchestration/modifications of orchestrated sounds with the ensemble

The interaction with the ensemble went very well. Being able to have direct feedback on the composed musical ideas and how to realize them is a privilege I wish I could have for every composition I write. Each musician was able to give to the composers their opinions on the feasibility and the efficiency of the musical materials. Whenever an aspect of the music didn't sound as planned, the musicians of the ensemble would suggest alternative solutions that would bring the sonority closer to the target sound. In short, I thought the interaction with the ensemble during the workshops was very useful and allowed me to learn a lot about orchestration techniques that I would have never learnt in the context of a traditional orchestration course.

Most modifications I had to make concerned pitch adjustments and timbral effects (through extended techniques). Because the musicians could hear the target sounds that the composers were trying to reproduce, they were able to come up with a few solutions to reproduce some of the timbral and spatial characteristics of these sounds that I would have never thought of.

III. Presentation of the Piece

a) General concept and program notes

I used light and photography techniques as an inspiration for creating different spaces in music. The title of the piece, Flou (blurry), flash, lucide (lucid) makes a direct reference to the effects and processes I used in the piece: an opposition between blurry and clear (lucid) materials, with the occurrence of some strong attacks/articulations representing the flashes (the first attack of the piece for instance).

Program notes:
With this piece, I intended to make a metaphor between photography and music, by drawing an analogy between light as used in photography, and space in music. Both can be used to create tension, emphasis on small details, a sense of separation or even to provide additional depth to the main materials. I was inspired from photography and light techniques, which allowed me to develop orchestration techniques that would directly influence our perception of space. Just like in photography, some musical materials in the piece can be clearly identified while others may be perceived as blurry and harder to grasp.

b) Form and structure of the piece

I created the general form of the piece based on three different light techniques that can be used in photography, and their metaphoric spatial meaning. The first section of the piece, from the beginning to measure 55, is based on two different photography techniques. The first one, called "tension," is meant to generate opposition and tension with space between elements of the story (in the same way that light in photography can generate opposition between characters or places in photographs). The second technique, called "emphasis" is made to lead the eyes (or the ears in this context) to what really matters, with lines, shapes, and negative space. The concept of negative space is a term used in the visual arts to describe the space surrounding an object. A negative space usually doesn't contain as many details as a positive space. This concept is similar to the concept of background and foreground in music.

Most of the opposition in this section is generated by the dichotomy between the first attack (measure 1) and its composed resonance, which both come from the orchestrated sound of the "pan" target sound. Some tension is also generated by breaking the expectation of the listeners (through a use of silences), by using "blurry" effects (small modifications of musical parameters in materials that have been heard previously in the piece), or by progressively building up the polyphony. The attack and the resonance (sometimes composed as a fast iteration of quiet pizzicato sounds, sometimes orchestrated in a more complex way with held notes and timbral effects) keep coming back in more complex forms throughout the section until they completely fuse (measure 32) and allow the perception of a completely different space (measure 39 to 55). This new space still uses the notion of tension and opposition that characterizes the first space of this section, but this time the contrasted materials are opposed in the same timespan (instead of being opposed one after the other like at the beginning of the piece): the frantic and chaotic noisy materials on the foreground are opposed to the background obsessive ascending melody at the piano that keeps oscillating between dynamics while getting louder and louder (and progressively come to the foreground). This space deals with the concept of negative space, as the attention of the listeners keeps alternating between a focus on the chaotic noise and a focus on the piano melody, as the latter constantly shifts between foreground and background.

The second section of the piece, from measure 56 to 84, is based on the technique of "substance" in photography. This technique aims to provide additional depth to the story by reinforcing the primary subject with light (or space in this piece). The beginning of the section starts by featuring the "Vibration F#" sound, to add a second element to the narration and bring another

change of space at the same time. At measure 69, a violent clash between the registers happens in the piano (a very loud chord using extreme registers is opposed to a quiet chord using a restricted register), which generates the sensation of a sudden change of space (from wide space to narrow space). This change of space also includes the reminiscence of the resonance of the pan sound previously heard in the first section of the piece. By the end of the section, the narrow space material has become louder and now precedes the wide space material.

The last section of the composition (measure 85 until the end) deals with the photography technique called "separation," which is meant to give shape to a story with basic tonality differentiation. The idea of this section was to start from silence and progressively extend the space by accumulation (of rhythms, lines, harmony, and timbre). At the beginning of this part comes a new material that was partly introduced at the end of the second section (from measure 79, in the violin part, then in other parts): an obsessive recurring small descending scale that continuously gets longer, quieter, and lower in pitch. The unfolding of this heterophonic texture progressively opens a wide space that gets filled by the last musical material of the piece: the "Sinus low vibration" sound. This final sound slowly disappears in the same space it opened, leaving a quiet never-ending resonance as an integral part of the space.

IV. Conclusion

When I wrote *Flou, flash, lucide*, the conception of space was mostly an abstract idea that related to the form and the musical materials of the piece. I realized during the rehearsals and the concert that a unified blend between the instruments was necessary for the listeners to be able to perceive the projected space in the music. Without this timbral blend, almost no composed space in the music could be perceptible. I was overall satisfied with the general perception of space and changes of space throughout the piece. Most composed spatial changes were effective (the most perceptible spatial effects were obtained through timbral augmentation and timbral resonance), and I noticed that some spatial changes were quite subjective and subtle in the piece (since space perception is the result of the subtle combination/mixing of dynamics, timbre, pitch, rhythm, and more parameters). Using materials that project contrasting spaces was very helpful to amplify the perception of space changes (like the dichotomy between the recurring first attack and its orchestrated resonance, for instance).

The SAT project was a very positive and fruitful experience that allowed me to expand my knowledge and my practice of orchestration related to space. The constant interaction with the musicians during the workshops was one of the most valuable and helpful aspects of this project. Indeed, being able to have direct feedback on the composed musical ideas and how to realize them is a privilege I wish I could have for every composition I write. Each musician was able to give to composers their opinions on the feasibility and the efficiency of the musical materials and parameters. If anything didn't work or didn't sound in the way I thought it would sound, the musicians of the ensemble would give me another solution that would work better and would reach a sonority even closer to the target sound than my original solution. Furthermore, I found that working with annotated parts between and during the workshops was quite useful as it allowed me to understand better how the instruments work (in terms of fingerings for instance, mutes, effects, etc.). It also made the correction process much faster as I could rely on the

annotated parts of the musicians on top of my own notes that I took during the workshops. It was a good way to ensure that I didn't forget anything that came up or was suggested by the musicians during the workshops.

To conclude, I thought the interaction with the ensemble during the workshops was very useful and allowed me to learn a lot about orchestration techniques that I would have never learnt in the context of a traditional orchestration course. Even though space has always been at the center of my musical composition practice (especially when composing electroacoustic or mixed music), the specific focus on space perception and space variations in this project allowed me to think about music and instrumental composition in new ways that I had not thought about before. I now consider space as an essential parameter resulting from timbral blend and timbral segregation when I write and listen to music. I will try to expand this practice in my next composition and research projects.

Appendix C

An evolving ecology— a conceptual approach to "space as timbre" Individual Report

Anqi Liu (University of California San Diego)

My primary motivation was to observe how a machine would interpret and transcribe these inherently organic and variable sonic phenomena into a structured orchestral format. Snow sounds, with their ever-changing dynamics and unpredictability, presented a fitting challenge. Would the software be able to capture the delicate crunches, the muffled silence, the soft patter of snowflakes, or the harsh winds that accompany a snowstorm?

To further complicate this, the concept of 'space' played an integral role in this experiment. Space, in terms of both physical space where the sounds originate and the metaphorical space within the composition, dictates the formal structure and progression of a musical piece. The question then arises - how would the software understand and interpret this 'space'? Would it be able to recognize the vast emptiness of a snowy field or the intimate closeness of snowflakes landing on a windowpane?

These questions took me on a fascinating journey, pushing both the limits of computer-aided software and my own understanding of sound, orchestration, and space. The results were intriguing, highlighting both the potentials and limitations of technology when faced with the complexities of sonic phenomena. The software captured the dynamic feature of the target sounds by suggesting busy rhythmic patterns interspersed with varied pitch combinations. Yet, one can't help but ponder: How does this algorithmic solution differ from stochastic choices? Moreover, the software's output demanded further optimization. It's vital to consider both the inherent capabilities of instruments and the physical limitations of human performers. On multiple occasions, the software proposed notes that were either beyond the playable range of

specific instruments or exceeded the feasible playing capabilities of a musician. Furthermore, refining the software's initial solution for notation proved labor-intensive. For reference, I've included both the original solution provided by the software and the post-processed score notation.

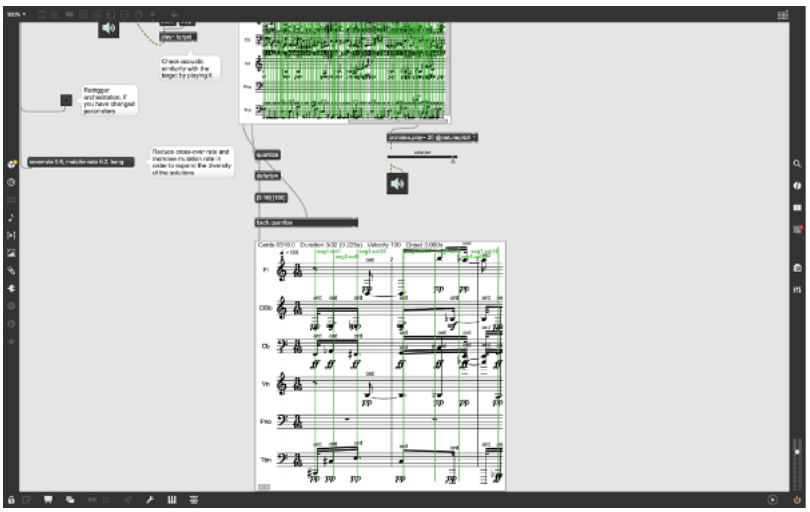


Figure 2: Suggested orchestration by Orchidea

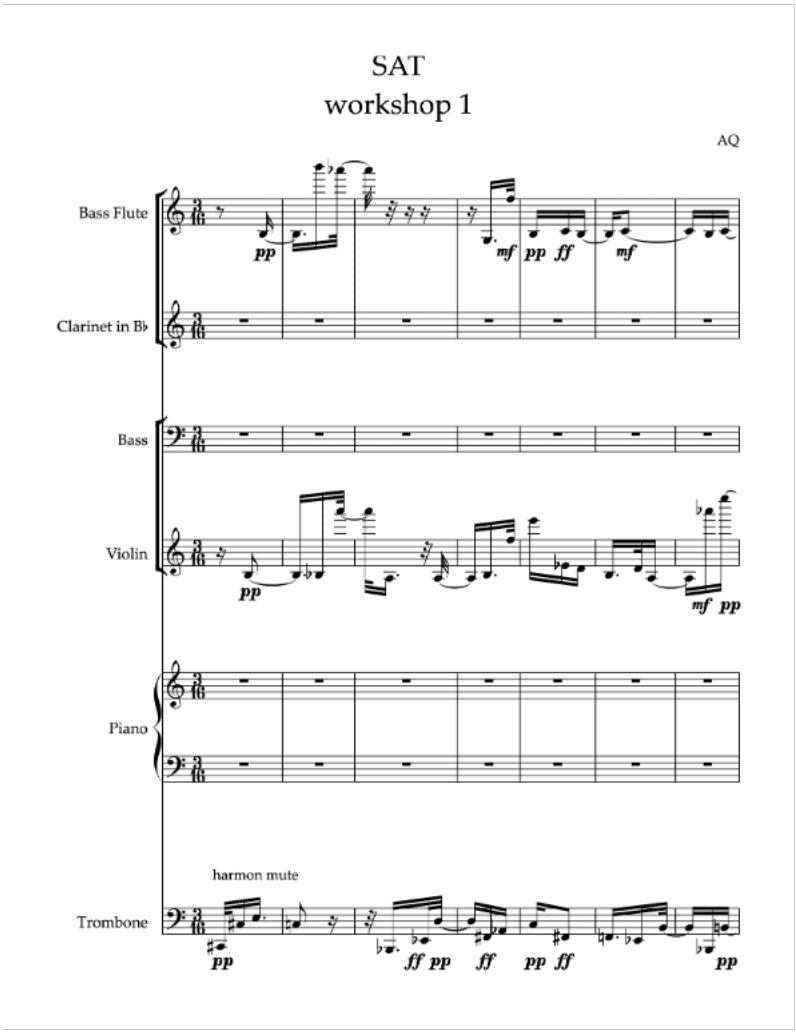


Figure 3: Anqi Liu's notation based on Orchidea suggestion

This endeavor fostered profound reflection. What nuances were lost or overlooked in the computer- aided orchestration process? Addressing this query is pivotal, given my years of exploration into the role of meticulously crafted scores that guide musicians understanding of our surrounding 'spaces' and 'ecologies'. I view past experiences, memories, training, and learnings as part of a psychological ecosystem influencing our cognitive patterns, decision-making, and behaviors. Yet, it was clear that the software did not integrate the human element into its orchestration suggestions. While the software's pitch and rhythm combinations were captivating, how could orchestration exist in isolation from the human nuances of playing an instrument, often acquired after years of rigorous training?

During the three enlightening workshops led by the fantastic musicians at McGill, it became clear that the software had certain limitations when it came to suggesting orchestration. But, it's essential to put this in context: I intentionally chose the snow sounds, knowing well that these nuanced, ephemeral sounds aren't necessarily what computer-aided software typically excels in.

My motivation, as previously mentioned, was to explore the boundaries of this computer-aided composition approach, and in doing so, understand its nuances and subtleties better. Many of my fellow composers achieved impressive results using the software, and their outcomes resonated well with the tool's capabilities. While my results took on a different hue, I too found satisfaction in this exploration. My journey was not just about achieving the perfect sonic representation but also about understanding the interplay between nature, technology, and human interpretation. The experience gave me invaluable insights into the balance between the structured precision of technology and the ever-changing dynamics of sonic phenomena. Embracing this tension, understanding it, and allowing it to inform my work has been an immensely rewarding aspect of this journey.

After these valuable experiences, my reflections took me down a different path. Instead of solely relying on the computer-aided software, I felt an urge to explore alternative solutions. This divergence wasn't just about moving away from technological means, but it also called into question the very linguistic model that has anchored musical endeavors for centuries: notation. Just as technology provides a framework to process and produce, so does musical notation offer a language for our sonic expressions. Both, at their heart, are linguistic models, shaping and codifying our interactions with sound.

Linguistic models, by their very nature, simplify and codify complex phenomena into understandable, repeatable formats. At their core, they seek to translate the inexpressible breadth of human experience into a structured framework. And while this provides us with a universally comprehensible language, it often truncates the richness of individual experiences. In music, traditional notation serves as a linguistic model, streamlining vast sonic landscapes into a series of symbols on a stave. But every symbol, every note penned, carries with it centuries of cultural, historical, and personal baggage, and may not truly encapsulate the essence of the sound or intangibility the musician wishes to convey. As the experimentation with the software revealed, while it could mimic patterns and suggest notations, it lacked the subtle understanding of human connection, gesture, and emotion. It's an experience that can't be boiled down to mere ones and zeroes.

In light of these revelations, I embarked on a quest to sculpt microscopic spaces within the gap of musical notation. A traditional score, while rich in its own right, is bound by its linear, time-bound representation of sound. My sculpture sought to break away from these confines, offering a more holistic, spatial, and interactive experience. By doing so, I hoped to create a system where the notation itself becomes an evolving entity, much like the ecologies we had discussed. The new notation would factor in the performer's experiences, memories, and training, enabling a evanescent rendition each time, thus bridging the gap between the rigidness of linguistic models and the fluidity of human experiences.

To further address the philosophy behind these sonic experiments, I want to share a human experience back to the Song Dynasty's (960-1279) Ru Porcelain. Artisans once tried, failed, and ultimately succeeded in capturing the ethereal blue of the sky on their creations. Yet, the true magic wasn't just in capturing that blue, but embracing the unpredictable alchemy of iron in the firing process. A challenge intensified by the capricious nature of the iron, making each

porcelain's final hue an unpredictable marvel. Even with the most diligent adherence to the ancient craftsman's techniques, the outcome remained an enigma until revealed, echoing the essence of real-time experimentation.

For instance, in the following description in the performance notes, I introduce the situation "threshold" described as in-between and ongoing. In this situation, uncertainties and unpredictabilities are unenviable. The thresholds are revealed in multiple parameters while playing the instruments. Part of the concepts of the piece is to find, explore and define those thresholds. The thresholds are conditions of uncertainties and are conditions of the edge of new or different things starting emerging; therefore it's always in the in-between and either-or situations.

Taking string instruments as examples, the representation of thresholds on the bow showcases the deliberate effort to merge the tactile experiences of a performer with the theoretical concepts of sound generation. When the bow interacts with the string, its positioning and pressure serve as parameters, defining the produced sound's tonal color, frequency, and amplitude. These parameters, however sophisticated, are bound by physicality. Yet, it's this very physicality that aids in conveying the intangible.

String instruments are concentrated on triggering two thresholds related to the bow:

- bow pressure- notated in the three-line stave named as **"Bow Pressure"** &
- bow position- notated in the three-line stave named as **"Bow Position"**

how to read the three-line stave in the "Bow Pressure" system:

- bottom line and under the bottom line: very heavy bow pressure
- mid line: thresholds (see below for more details)
- top line: very light bow pressure

how to read the three-line stave in the "Bow Position" system:

- bottom line and under the bottom line: on the bridge and underneath the bridge
- mid line: thresholds (see below with more details)
- top line: on the finger board
- above the top line: further on the finger board

When symbols ■ ▶ appear, it means thresholds related to the notated parameters are triggered. They always followed by a horizontal line as notated meaning that the thresholds situations are suspended and hold for a bit of time.

- thresholds ■ in the "Bow Pressure" system:

The threshold situation regarding the bow pressure is those spots where the pitch being bent. The threshold in this circumstance is where the pitch starts distorting while applying pressures.

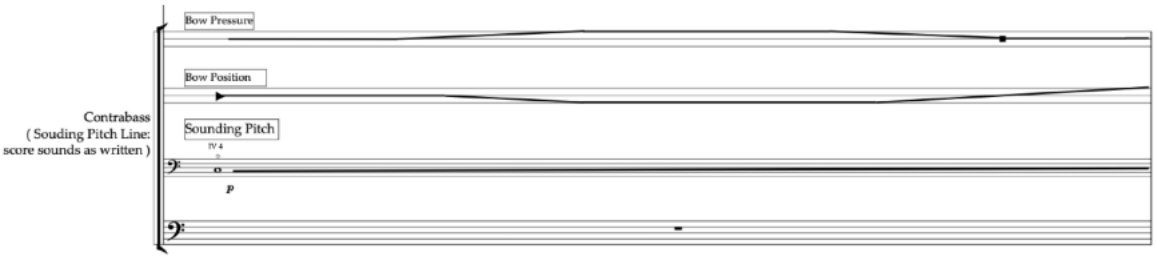
interpretation of the thresholds with the three-line stave in the bow pressure system:
It always positioned in the mid of the three-line stave of the Bow Pressure system.

- thresholds ▶ in the "Bow Position" system:

while moving your bow in between the fingerboard and the bridge with simultaneously trying to produce the given high or normal particles, there is a spot where the frequency of the partial emerges from a rather hazy, noise-based texture. In other words, the closer to the bridge the more focused the pitch. The threshold in this particular situation is where the pitch begins becoming hazy and shadowed; or vice versa, the threshold is where the pitch starts emerging from the rather vague sonic texture.

interpretation of the thresholds with the three-line stave in the bow position system:
It always positioned in the mid line of the three-line stave of the Bow Position system.

Score Example:



In this example, at m. 1 of the contrabass, let's look at the bow position system. The notation indicates that starting with the threshold of the showed/hazy 4th partial on the C-string, after holding on the threshold, the bow gradually moves to the bridge and holds for a bit of time, and then continues moving to the fingerboard.

In the description above, we realize the bow positions constantly change. The only exception is when the horizontal line appears. In the situation of the horizontal line, the notated situation needs to be held for a bit of time. As mentioned before, the threshold situation always followed by the horizontal line, which the threshold situation is always hold for a bit of time.

Please also remember that the degree of *sul tasto* is unlimited, which means you can decide how far you want to travel on the fingerboard if the indication is above the top line in the Bow Position system.

The above mentioned phenomenon is applied to the "Bow Pressure" system.

Figure 4: Anqi Liu, While Snow Buries*…, pp. 2-3*

This approach to orchestration and notation isn't about replacing traditional systems but enriching them. It's a manifestation of the idea that, while computational models can rapidly process vast amounts of information, they often skim over the deeper, intricate layers of human connection. For example, with Ru Porcelain, while the capricious nature of the iron needed to be tamed to a certain extent to achieve the desired hue of blue, it still retained its inherent unpredictability.

In contrast, my scoring system embraces the unpredictable, allowing it to flow freely, shaped by a multitude of human phenomena. The essence of the score changes when each performer brings their personal touch, be it through subtle differences in interpretation, fleeting moments of subconscious interference, or even a rekindling of past memories evoked during the performance. These moments, rich with individual experience and unspoken narratives, can influence the micro gestures utilized to approach these thresholds. Instead of being precise and defined, the application of technique becomes a reflection of these underlying psychological ecologies. Where one might have been sure of a particular bowing situation, a sudden rush of surrounding sounds might cause the hand to waver, deviating from the exact point of contact on the string, or altering the pressure applied. These nuances, subtle yet profoundly impactful, can mean the difference between hitting the threshold, fleetingly passing it, or perhaps not reaching it at all.

Unlike the exacting requirements of firing Ru Porcelain, where specific conditions had to be met for success, in this framework, these unpredictabilities aren't regarded as inconsistencies or errors; rather, they are celebrated as integral components of the sonic palette. The process of achieving or missing the threshold becomes fluid, imbued with the weight of human experience and memory. Instead of a singular, virtuosic rendition, what emerges is a flux of sounds. It's less about a binary distinction between structure and spontaneity, and more about fostering a dynamic symbiosis where the two coexist, allowing the musical narrative to be constantly reshaped by individual experience and memory.

It is this confluence of the tangible and intangible that sets this approach apart. Technique, no longer rigid and fixed, becomes a living, breathing entity, evolving with each performance. The sonic results, therefore, are not just a product of practiced gestures but also a testament to the inner life of the performer.

This approach to orchestration and notation isn't about replacing traditional systems but augmenting them. With these new systems and my continued exploration, I aim to foster a space where technology aids human experiences without stifling it. Where linguistic models serve as foundational structures, yet are malleable enough to accommodate the unpredictability of human experiences. This journey has become an embodiment of my philosophical musings on space, ecology, human connection, and the delicate balance between structure and uncertainties. It's a call to recognize that while technology can process vast data rapidly, it often skims the intricacies of human connection. I urge musicians, technologists, and art enthusiasts to see this as an opportunity – to embrace both the structure of linguistic models and the unpredictability of human experience.

References

- Cimini, Amy. (2022). *Wild Sound Maryanne: Amacher and the tenses of audible life*. Oxford University Press
- Cella, Carmine Emanuele. (2022). "Orchidea: A Comprehensive Framework for Target-Based Computer- Assisted Dynamic Orchestration." *Journal of New Music Research* 51, no. 1: 40–68. https://doi.org/10.1080/09298215.2022.2150650.
- Leman (Eds.). (2010). *Musical gestures: Sound, movement, and meaning.* Routledge.
- Kaufmann, Walter. (1975). *Tibetan Buddhist Chant: Musical Notations and Interpretations of a Song Book by the Bkah Brgyud Pa and Sa Skyapa Sects.* Indiana University humanities series.
- Radigue, Eliane, and Julia Eckhardt. (2019). *Intermediary Spaces*. Umland: Umland editions,
- Takemitsu, Toru, Yoshiko Kakudo, and Glenn Glasow. (1995). *Confronting Silence: Selected Writings*. Landham, MD: Scarecrow Press.
- Wittgenstein, Ludwig. (1992). *Tractatus Logico-Philosophicus*. Atlantic Highlands, NJ: Humanities Press International.

[i] An impulse response is an imprint of the acoustic characteristics of a given space. Impuse responses provide information about the reverberant characteristics of a room and can be used to modulate a given sound by means of convolution. What Is Impulse Response In Audio ? (n.d.). SoundAssured. https://www.soundassured.com/blogs/blog/what-is-impulse-response-in-audio

[ii] Target-based computer-assisted orchestration is the process of searching for combinations of orchestral sounds in a database to replicate a specific sound, referred to as the target. (Evripidis Bampis et al., "Target-based Computer-assisted Orchestration: Complexity and Approximation Algorithms," European Journal of Operational Research 304, no. 3 (2023): 926-938.)

[iii] Orchidea is a framework for static (for sounds that don't change in time) and dynamic (for sounds that evolve in time) computer-assisted orchestration. (Ircam, HEM, UC Berkeley, and Carmine-Emanuele Cella, "Orchidea: Creative Computer-Assisted Orchestration," accessed June 11, 2024, http://www.orch-idea.org/.)

[iv] Orchidea contains three main datasets: the TinySol dataset (a collection of orchestral instruments in Ordinario playing style), OrchideaSol (a large collection of orchestral instruments in several playing styles), and FullSol (a very large collection of orchestral instruments in several playing styles, including extended techniques).

[v] More information about the parameters of Orchidea can be found in the Orchidea tutorials under the TOR modules of Timbre and Orchestration Resources. (https://timbreandorchestration.org/tor/modules/assisted-orchestration/orchidea/tutorials)

[vi] Regardless of what the sound referent might be, the illusion may not be perfect enough for the listener to recognize the reference. Despite this, source content awareness can help a listener perceive the illusion of the referential dimension of a resynthesized sound. A sound can retain certain essential features that define its identity even if identity is not the primary concern through the resynthesis process.

View publication stats